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Machine learning for risk profiling: An analysis of pension fund participants

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ABSTRACT

This study examines the use of machine learning (ML) techniques for profiling the risk of pension fund participants. We analyze a dataset of 81,563 individual investors in a major Turkish pension fund company (2018–2022), comparing various ML models to the regulatory benchmark. Using recursive feature elimination, we identify self-reported risk attitudes and age – with a nonlinear relationship – as the most important predictors of actual portfolio risk. Our cross-validation results indicate that boosting methods yield modest improvements in predictive accuracy relative to the regulatory risk score. Notably, the performance from using just four variables is comparable to that from using the full questionnaire. Although the overall explanatory power remains modest across all models (R^2 of 0.13–0.17), the findings suggest that ML can enhance risk profiling by identifying informative variables and capturing nonlinear relationships. These results have practical implications for designing more efficient risk assessment tools in pension fund settings, potentially simplifying questionnaires without sacrificing predictive accuracy.

1. Introduction

The alignment between investors' risk preferences and their portfolio choices represents a fundamental challenge in financial markets. Mismatches between stated preferences and actual investment behavior can lead to suboptimal outcomes, including premature liquidation during market downturns, inadequate retirement savings, and reduced long-term wealth accumulation (Brayman et al., 2015). This challenge has become increasingly critical as pension systems worldwide shift toward defined contribution plans, placing greater responsibility on individual participants to make appropriate investment decisions. In Türkiye, over 9 million participants managed approximately USD 55–60 billion in pension assets as of 2024, making accurate risk assessment essential for both individual welfare and financial system stability (Pension Monitoring Center of Türkiye (EGM), 2025).

This regulatory requirement, however, is coupled with a significant operational challenge. Pension funds and financial institutions, often interacting with clients via high-friction channels like phone or digital apps, face major hurdles with long, complex surveys. This "participant friction" can lead to low completion rates and incomplete data (Brayman et al., 2015), creating a critical tension between regulatory compliance and operational efficiency. Furthermore, studies examining global risk assessment practices have identified important limitations in many existing questionnaires. Roszkowski and Grable (2005) and Bouchey (2004) note that many questionnaires fail to meet established psychometric standards and often show limited ability to predict actual investment behavior.

Risk preferences are commonly assessed through either incentivized experimental methods that reveal preferences through observable choices, or self-report measures that elicit stated preferences directly. The literature presents an ongoing debate regarding which approach better captures true risk preferences (Friedman et al., 2014; Beshears et al., 2008; Pedroni et al., 2017; Mata et al., 2018). Some studies suggest that revealed and stated preferences show limited consistency across methods (Pedroni et al., 2017). However, Mata et al. (2018) note that revealed preference measures do not necessarily demonstrate superior predictive validity compared to stated preference measures. Nevertheless, both types of risk assessment approaches typically exhibit modest predictive power for actual financial risk-taking behavior, with generally low R^2 values (Barsky et al., 1997; Dohmen et al., 2011; Beauchamp et al., 2017; Gürdal et al., 2017; Kapteyn & Teppa, 2011).

Despite growing recognition of these limitations, three critical research gaps remain. First, while studies have documented the modest predictive power of risk questionnaires (Beauchamp et al., 2017; Dohmen et al., 2011), limited research has systematically compared machine learning (ML) approaches against regulatory benchmarks using actual portfolio data in emerging market contexts. Second, the

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relative importance of different questionnaire items for predictive accuracy remains unclear, with no consensus on whether comprehensive assessments are necessary or if simplified tools could achieve comparable results—a question with significant practical implications for the industry's participant friction problem. Third, the potential for non-linear relationships between demographic factors and risk-taking behavior has received insufficient attention, particularly in emerging markets where lifecycle patterns may differ.

This study contributes to this line of research by addressing these three gaps directly. First, we systematically compare a suite of ML models against the regulatory benchmark, using a large, real-world dataset of 81,563 investors' actual portfolio risk from an emerging market. Second, we apply feature selection methods to identify a concise, 4-variable set of predictors, directly addressing the trade-off between accuracy and the industry's need to reduce operational friction. This shorter set of inputs offers a practical advantage for large-scale implementation, especially in phone-based or digital enrollment channels where lengthy questionnaires create participation barriers. Third, we explicitly model and confirm a non-linear relationship between age and risk-taking. Our work moves beyond the Adekunle et al. (2023) study (which predicts survey-based risk) and the Kuzubaş and Saltoğlu (2024) factor analysis by providing a practical, data-driven path toward a more efficient and operationally viable assessment tool.

Our work is situated within a rapidly evolving literature on ML in finance. While foundational work focused on credit risk, bankruptcy prediction, and fraud detection (e.g., Khandani et al., 2010; Barboza et al., 2017; Shi et al., 2022), recent research has expanded into three distinct areas highly relevant to our study.

The first area is the application of ML to personalization in wealth management, particularly in the high-growth field of robo-advising. This research directly intersects with behavioral finance, as studies now evaluate how these technologies address (or fail to address) investor rationality, risk tolerance, and expectations (Eichler & Schwab, 2024). Concurrently, researchers are developing complex deep learning frameworks for risk-aligned portfolio optimization (Nguyen, 2025). This trend is particularly relevant for our work, as industry analysis now highlights the central role of AI in transforming pension systems and governance (Hayman & Genevieve, 2024), though significant concerns remain about model-fit and the risks of using historical data for long-term predictions (Benefits Canada, 2025).

The second stream of research is exploring alternative data, with a strong focus on using Natural Language Processing (NLP) to extract investor sentiment from unstructured text. This field has matured rapidly, with recent systematic reviews mapping its evolution (Huynh et al., 2025). Current applications range from general sentiment analysis (Hansen & Borch, 2022) to specific tasks like real-time, high-frequency stock return prediction (Cai et al., 2024) and even using the text from lottery-choice scenarios to risk-profile investors (Thomas et al., 2023). This research path, focused on mining new data sources, contrasts with our study's focus on making existing, mandatory survey data smarter and more efficient.

Finally, the adoption of these increasingly complex models has created a significant regulatory and practical challenge: the black box'' problem. This is no longer a niche concern; a wave of recent systematic reviews has highlighted the black box'' as a central academic and industry-wide challenge (e.g., Khan et al., 2025; Černevičien˙ e & Kabašinskas, 2024). This concern is mirrored by stakeholders and regulators, with major bodies like the CFA Institute (Wilson, 2025) and the Bank of England Gharbawi et al. (2024) now issuing guidance on AI transparency. This has spurred a critical field of Explainable AI (XAI) focused on building transparent models for parallel applications. This includes developing interpretable models for systemic risk (Tang et al., 2024), creating frameworks to balance the accuracy-interpretability trade-off (Hamerle et al., 2024; Mena et al., 2024), and applying XAI techniques to fraud detection (Yaseen & Al-Amarneh, 2025) and credit risk (Nallakaruppan et al., 2024). Recent systematic reviews on default prediction further confirm the move toward hybrid and explainable models for regulatory acceptance (Reisen et al., 2024). Our study contributes directly to this XAI conversation. By demonstrating that an inherently interpretable, 4-variable model can achieve robust performance, we provide a practical solution that balances predictive power with the critical, and often overlooked, needs for operational efficiency and regulatory transparency.

The remainder of the paper is organized as follows: Section 2 provides a summary of the risk profiling in the private pension system in Türkiye. Section 3 presents the data set and empirical analysis of the questionnaire. Finally, Section 4 concludes with a summary and discussion of the results. The questions in the risk questionnaire and the details on the machine learning models used in the analysis are provided in the Appendix.

2. Research context and data

2.1. Türkiye's private pension system and risk profiling requirements

Türkiye's private pension system serves as a supplement to the state-funded social security retirement system. Established through legislation in 2001 and fully operational by 2003 with six pension companies, the system saw a total asset investment of approximately 20 billion USD in 2022, representing around 3% of the nation's GDP. This system operates on a defined contribution basis with voluntary participation, allowing individuals to invest in mutual funds managed by asset management firms. Unlike the US pension system, participants cannot invest directly in individual stocks but can choose from mutual funds that invest in equities, bonds, or commodities. In 2017, the Capital Markets Board of Türkiye (CMB) mandated that financial institutions evaluate the risk tolerance of their clients and provide financial advice that aligns with clients' risk preferences. This mandate, known as suitability, requires financial advisors to collect relevant information about clients' investment goals and risk attitudes before making portfolio recommendations, aligning with the EU's Markets in Financial Instruments Directive (MiFID). This regulatory framework provides a systematic setting for examining risk profiling approaches, as all pension companies are required to use standardized assessment tools while participants' actual portfolio choices may serve as a measure of risk-taking behavior.

2.2. The CMB risk assessment framework

To facilitate compliance with the suitability requirement, the CMB developed a standardized risk questionnaire designed to classify clients according to their risk attitudes. The questionnaire consists of eight questions[1]; the first seven assess risk-taking behavior across different dimensions, while the eighth concerns sensitivity to interest-bearing assets and is excluded from the total risk score.; the first seven assess risk-taking behavior across different dimensions, while the eighth concerns sensitivity to interest-bearing assets and is excluded from the total risk score. Each participant's risk profile is determined by weighted scores from the seven relevant questions, which categorize them into one of four risk groups: low (0–15 points), medium (16–23 points), high (24–32 points), and very high risk (33–47 points).

While this standardized approach ensures consistency across institutions, studies examining global risk assessment practices have identified potential limitations in existing questionnaires. Brayman et al. (2015) report that only a small percentage of risk assessments may adequately fulfill their intended purpose, with issues including suboptimal question design and simplistic scoring approaches. Several researchers have

[1] The complete questionnaire with all questions, response options, and scoring weights is provided in Appendix A1.

noted that many questionnaires used in practice may not meet established psychometric standards and often show limited ability to predict actual investment behavior (Bouchey, 2004; Roszkowski & Grable, 2005).

These observations regarding current risk assessment practices, combined with developments in machine learning techniques, suggest potential opportunities to examine risk profiling effectiveness. Machine learning approaches may offer the ability to identify informative questions, account for non-linear relationships between variables, and potentially provide improved predictions of risk-taking behavior. This raises the question of whether machine learning techniques might enhance the current regulatory approach to risk profiling in Türkiye's pension system.

2.3. Research setting and data overview

This study examines these questions using data from a major pension fund company in Türkiye, covering the years 2018 to 2022 and comprising information on 81,563 participants. The scale of this dataset may be suitable for machine learning applications, while the regulatory requirement for standardized questionnaires provides consistency in risk assessment across participants.

Our analysis is cross-sectional; we observe each participant's risk questionnaire responses at a single point in time, along with their demographic characteristics and average portfolio risk over the 2018–2022 period. This allows us to examine the relationship between stated preferences (questionnaire responses) and revealed preferences (actual portfolio choices), which represents an important consideration in risk profiling research.

Several aspects of Türkiye's pension system may provide advantages for this research. First, the mandatory use of the CMB questionnaire ensures standardized risk assessment across participants. Second, the mutual fund structure of investments provides a measurable indicator of portfolio risk based on fund volatility ratings. Third, the regulatory environment creates incentives for accurate risk assessment, which may make our findings relevant to policy and practice.

This research setting allows us to examine three questions that may have implications for risk profiling in pension systems: (1) Whether machine learning models can improve predictive accuracy compared to the current regulatory approach; (2) Which variables appear most informative for predicting actual risk-taking behavior; and (3) Whether risk assessment questionnaires might be simplified without substantially reducing predictive accuracy. The empirical analysis that follows attempts to provide evidence on these questions, which may inform regulatory policy and industry practice.

3. Data and empirical analysis

This section details our empirical methodology, structured to proceed from data exploration to model evaluation. We begin by defining our dependent variable, portfolio risk. We then introduce our predictor variables and analyze descriptive patterns, highlighting relationships that inform our modeling strategy. Following this, we outline our modeling approach and employ recursive feature elimination (RFE) to identify the most salient predictors. Finally, we present a comparison of machine learning models using cross-validation and conclude with an interpretation of the results' practical significance.

The data, covering 81,563 participants from 2018–2022, include responses to the CMB risk assessment questionnaire, demographic information, and portfolio risk measures derived from actual investment allocations. Table 1 shows the sample distribution, indicating that most participants are male, married, college graduates, with an average age of 39.48 years.

3.1. Portfolio risk: The dependent variable

In the Turkish pension fund system, participants choose from mutual funds that invest across asset classes. The risk level of each fund is classified on a scale from 1 (lowest risk) to 7 (highest risk), determined by its annualized weekly return volatility over the past five years. The regulator provides the standardized formula for this calculation:

$$\sigma_f = \sqrt{\frac{m}{T-1}\sum_{t=1}^{T}(r_{f,t}-\bar{r})^2}$$

where $m = 52$, $T = 260$, $r_{f,t}$ is the weekly return of the fund, and $\bar{r}$ is its average weekly return over the five-year period. This standardized volatility measure allows us to construct a continuous portfolio risk variable that serves as our dependent variable across all model specifications (detailed in Appendix A2).

For each individual, we calculate their overall portfolio risk as the weighted average of the risk scores of their chosen funds. Our analysis uses the average portfolio risk for each participant over the 2018–2022 period. This continuous measure of revealed risk preference is similar to metrics used in prior literature, such as Beauchamp et al. (2017) and Cesarini et al. (2010), and offers greater granularity than binary indicators like stock ownership.

3.2. Predictor variables and descriptive patterns

Our predictor variables include participant responses to the seven core questions of the CMB risk survey alongside demographic and economic information. Here, we examine the relationships between these variables and portfolio risk to identify patterns that can inform our modeling strategy.

Table 2 presents the survey responses and portfolio risk by education, marital status, and gender. The investment horizon question (Q1) reveals that participants with higher education levels report shorter investment horizons, while married and male participants indicate longer horizons. Self-assessed financial literacy (Q2) demonstrates a negative association with education level but is positively associated with being male and married. Self-reported risk attitudes (Q3) and preferences in investment scenarios (Q4) both exhibit positive associations with education level. Similarly, married and male participants demonstrate higher risk tolerance in their responses. Responses to the loss reaction question (Q5) follow comparable patterns. Questions examining financial situations (Q6 and Q7) suggest that financial stability increases with education level and is generally higher among married and male participants. The observed portfolio risk measurements follow patterns similar to the questionnaire responses, increasing with education level and showing modest elevation for married and male participants.

Table 3 shows questionnaire responses and portfolio risk across age categories. An important finding is the non-linear relationship between age and risk attitudes, with risk-taking behavior increasing until middle age (35–44 category) before declining in older age groups. This pattern is particularly evident in portfolio risk measures, which reach 5.02 in the 35–44 age group before decreasing to 4.63 for participants aged 55 and older. This observed pattern is consistent with life-cycle investment theories. This observed non-linear relationship has important implications for our modeling approach. Linear models require explicit specification of polynomial terms to capture such patterns, while tree-based methods can identify these relationships through their recursive partitioning process.

Table 4 extends the analysis to income and employment sector. The data suggest that risk tolerance generally increases with income levels, and that public sector employees exhibit somewhat different response patterns compared to those in the private sector.

These descriptive findings provide motivation for our empirical approach. The non-linear age effect suggests that models capable of capturing such relationships – either through explicit feature engineering (e.g., adding an age-squared term) or inherently (e.g., tree-based

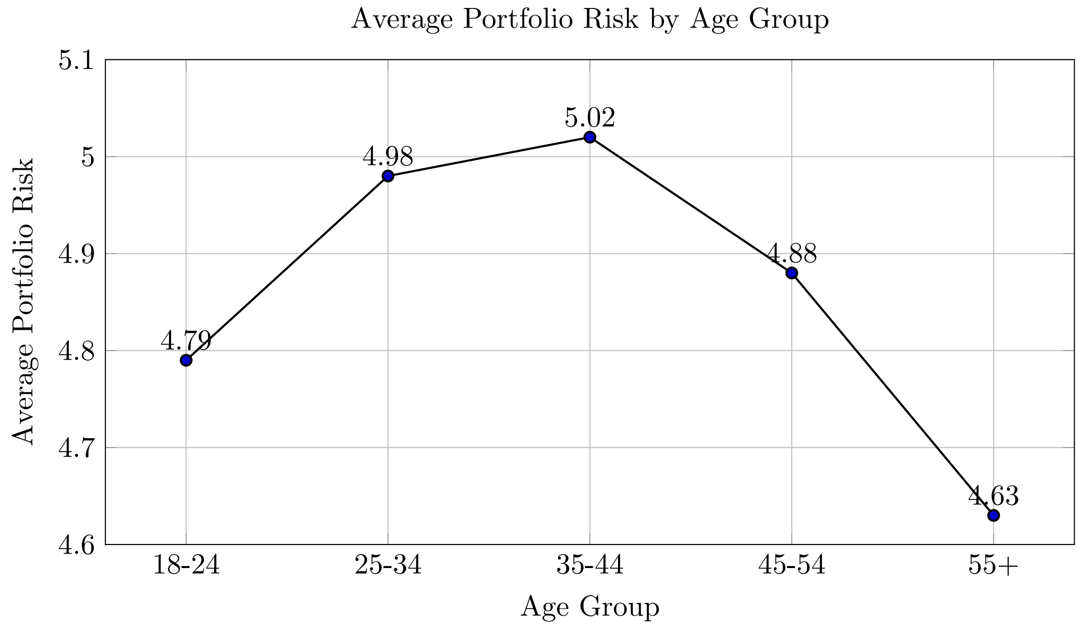

Fig. 1. Relationship between age and average portfolio risk.

Table 1
Demographic characteristics.

	College	Post-College	High School	Primary	Married	Male	Female	Age
Mean	0.57	0.07	0.31	0.06	0.73	0.68	0.32	39.48
S.D.	0.50	0.25	0.46	0.23	0.45	0.47	0.47	9.49
Min	0	0	0	0	0	0	0	18
Max	1	1	1	1	1	1	1	78.5
N.of Obs.	46 231	5712	24 954	4666	59 870	55 248	26 315	81 563

Notes: Demographic characteristics of the pension participants. There are four education categories: primary school, high school, college, and post-college. “Married” is a dummy variable equal to 1 if the participant is married, and “Male” is a dummy variable equal to 1 if the participant is male.

Table 2
Responses to the risk questionnaire and portfolio risk: Demographic characteristics.

	Post-Col.	College	High school	Primary	Married	Single	Male	Female
Q1	3.11	3.04	2.86	2.68	3.02	2.85	3.01	2.89
Q2	2.74	2.44	2.11	1.96	2.37	2.24	2.40	2.18
Q3	3.37	3.20	2.91	2.77	3.14	2.99	3.18	2.91
Q4	2.76	2.64	2.45	2.35	2.61	2.48	2.67	2.38
Q5	2.89	2.83	2.74	2.67	2.82	2.75	2.87	2.65
Q6	2.82	2.59	2.41	2.39	2.57	2.48	2.62	2.52
Q7	3.67	3.41	3.08	2.93	3.35	3.17	3.37	3.15
Portfolio risk	5.01	4.99	4.88	4.77	4.96	4.89	4.98	4.88

Notes: Descriptive statistics for risk survey questions, demographic characteristics, and the portfolio risk measure.

ensembles) – may offer better performance. The patterns associated with self-reported risk attitudes (Q3) and lottery choices (Q4) also suggest these items may be informative predictors.

3.3. Modeling approach and feature selection

To identify the most predictive variables, we apply the RFE algorithm. This method iteratively ranks and prunes features based on their contribution to model performance across our models: ordinary least squares (OLS), ridge, Lasso, CART, random forest, and three boosting algorithms (GBoost, XGBoost, and LightGBM). The equations and algorithmic details for each model are in Appendix A2. Categorical predictor variables – including education, marital status, gender, income quartiles, and employment sector – are transformed into binary variables using one-hot encoding to ensure consistent feature

Table 3
Responses to the questionnaire and portfolio risk: Age groups.

Age group	18–24	25–34	35–44	45–54	55+
Q1	2.54	2.99	3.17	2.85	2.35
Q2	2.08	2.24	2.36	2.44	2.45
Q3	2.68	3.01	3.17	3.18	3.07
Q4	2.29	2.49	2.62	2.66	2.6
Q5	2.67	2.75	2.83	2.84	2.79
Q6	2.28	2.39	2.55	2.69	2.83
Q7	2.78	3.16	3.33	3.47	3.57
Portfolio Risk	4.79	4.98	5.02	4.88	4.63

Notes: Average responses for risk survey questions and portfolio risk for different age categories.

Table 4
Responses to the questionnaire and portfolio risk: Income and sector.

	Income Q1	Income Q2	Income Q3	Income Q4	Public	Private
Q1	2.92	2.95	3.00	3.02	3.08	2.98
Q2	2.31	2.18	2.33	2.52	2.29	2.33
Q3	2.99	2.99	3.13	3.29	3.14	3.08
Q4	2.47	2.47	2.60	2.78	2.64	2.54
Q5	2.74	2.73	2.81	2.92	2.80	2.79
Q6	2.49	2.43	2.55	2.72	2.53	2.49
Q7	3.03	3.15	3.36	3.67	3.41	3.25
Portfolio Risk	4.93	4.91	4.95	5.02	5.06	4.94

Notes: Average responses for risk survey questions and portfolio risk for income quantiles (Q1=lowest, Q4=highest) and sectors of employment.

Table 5
Recursive feature selection.

	OLS	Random tree	Random forest	GBoost	XGBoost	LGBoost	Ridge	Lasso	Selection
Q1	4	2	15	2	2	3	4	3	**0**
Q2	3	4	14	4	3	2	3	4	**0**
Q3	1	1	1	1	1	4	1	1	**7**
Q4	2	1	1	1	1	6	2	2	**4**
Q5	5	5	12	5	5	8	5	5	**0**
Q6	7	12	11	9	7	7	7	7	**0**
Q7	8	11	10	7	8	5	8	9	**0**
Age	1	3	8	1	1	1	1	1	**6**
Age-Squared	1	1	7	3	14	14	1	1	**4**
Male	10	10	9	10	11	10	10	10	**0**
Married	9	14	1	11	9	11	9	8	**1**
High-School	14	7	4	13	13	13	14	13	**0**
College	13	9	6	12	10	12	13	14	**0**
Post-College	12	8	5	14	12	15	12	12	**0**
Income	11	6	13	6	6	1	11	11	**1**
Public	6	13	2	8	4	9	6	6	**0**

Notes: This table lists the feature rankings from the RFE process. A lower rank indicates greater importance. The Selection column shows the number of models in which a feature ranks first.

representation across all models. To accommodate the nonlinear age effect identified in our descriptive analysis, we include age-squared as a predictor in all linear models (OLS, ridge, Lasso, elastic net). Tree-based models (CART, random forest, and boosting algorithms) inherently capture these nonlinearities without requiring polynomial feature engineering.

All models are tuned using grid search with fivefold cross-validation on the training set (80% of data), with mean squared error (MSE) as the selection criterion. For example, random forest hyperparameters include the number of trees ($n\_estimators \in \{100, 200\}$) and maximum tree depth ($max\_depth \in \{5, 10\}$), with optimal values of 200 and 10, respectively. Complete details on the search spaces and optimal parameters for all models are in Appendix A3.

Table 5 lists the feature rankings from each model. The results are consistent with the patterns observed in our descriptive analysis. Self-reported risk attitude (Question 3) ranks as the most important predictor by seven of the eight models, aligning with prior work by Gürdal et al. (2017). Consistent with the life-cycle pattern shown in Table 3, age and age-squared are also highly important variables. The hypothetical lottery trade-off (Question 4) emerges as another key predictor. However, several other demographic and financial condition variables are less influential. These findings suggest that a small set, consisting of four variables – Questions 3 and 4, age, and age-squared – can capture a large proportion of the explanatory power.

To ensure the robustness of our feature selection findings, we conduct a comprehensive stability analysis across all 20 cross-validation folds (detailed in Appendix A4). For each fold, we apply the RFE algorithm to the training data, generating 160 total feature rankings (20 folds × 8 models) for each variable. Our stability analysis confirms that the four core features – Question 3 (self-reported risk attitude), Question 4 (lottery choice), age, and age-squared – consistently rank among the top predictors across all models and folds, with standard deviations below 5.0, which indicates that the selection is robust. Question 3, in particular, demonstrates exceptional stability (mean rank = 2.09, SD = 0.90), whereas all the other features have mean ranks of more than 5.0, with no first-place rankings. This cross-validation stability suggests that the concise feature set that we identify reflects consistent patterns in the data, rather than being highly sensitive to particular sampling variations.

3.4. Model performance and comparison

We assess model performance using 20-fold cross-validation, comparing models trained on both the full feature set and the smaller subset of four key features identified by RFE. We report the mean and standard deviation of multiple performance metrics: R^2, mean squared error (MSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). Complete cross-validation results for all metrics are in Tables A4–A19 (Appendix A5).

Table 6
Model comparison with cross-validation: All features.

Model	R^2-Mean	R^2-Std	MSE-Mean	MSE-Std	MAE-Mean	MAE-Std
Regulator	0.1314	0.0098	0.6851	0.01684	0.6402	0.0086
OLS	0.1546	0.0094	0.6668	0.0175	0.6300	0.0089
KNN	0.1349	0.0100	0.6824	0.0161	0.6388	0.0084
CART	0.1385	0.0093	0.6795	0.0157	0.6388	0.0081
ANN	0.1510	0.0109	0.6693	0.0158	0.6305	0.0086
Random Forest	0.1522	0.0093	0.6690	0.0157	0.6324	0.0083
GBoost	0.1659	0.0092	0.6579	0.0162	0.6252	0.0084
XGBoost	0.1608	0.0091	0.6619	0.0151	0.6273	0.0080
LGBoost	0.1672	0.0089	0.6569	0.0162	0.6247	0.0083
Ridge	0.1546	0.0094	0.6668	0.0175	0.6300	0.0089
Lasso	0.1538	0.0090	0.6675	0.0173	0.6306	0.0089
Elastic Net	0.1546	0.0094	0.6668	0.0175	0.6300	0.0089

Notes: This table compares the mean and standard deviation of R^2, MSE, and MAE values across different models using 20-fold cross-validation.

Table 7
Model comparison with cross-validation: Selected features.

Model	R^2-Mean	R^2-Std	MSE-Mean	MSE-Std	MAE-Mean	MAE-Std
Regulator	0.1314	0.0098	0.6851	0.01684	0.6402	0.0086
OLS	0.1343	0.0087	0.6829	0.0179	0.6409	0.0092
KNN	0.1159	0.0100	0.6974	0.0180	0.6484	0.0091
CART	0.1319	0.0077	0.6847	0.0162	0.6417	0.0086
ANN	0.1354	0.0086	0.6813	0.0169	0.6406	0.0094
Random Forest	0.1384	0.0076	0.6796	0.0167	0.6390	0.0086
GBoost	0.1433	0.0079	0.6758	0.0166	0.6376	0.0089
XGBoost	0.1387	0.0070	0.6794	0.0163	0.6395	0.0087
LGBoost	0.1442	0.0077	0.6750	0.0165	0.6371	0.0088
Ridge	0.1343	0.0087	0.6829	0.0179	0.6409	0.0092
Lasso	0.1326	0.0083	0.6842	0.0178	0.6416	0.0092
Elastic Net	0.1343	0.0086	0.6829	0.0179	0.6409	0.0092

Notes: This table compares model performance using the selected features (Question 3, Question 4, age, and age-squared).

As shown in Table 6, when all features are used, boosting methods (LightGBM, GBoost, XGBoost) outperform the regulatory benchmark and traditional linear models. LightGBM achieves an average R^2 of 0.1672, whereas the benchmark is 0.1314. The superior performance of these models may be due in part to their ability to capture the nonlinear age-related patterns observed in Fig. 1. Fig. 2 illustrates these results, showing that the ML models, in particular the boosting methods, have less performance variability across validation folds than the regulatory benchmark.

Next, we evaluate the models using only the four features selected. The results, summarized in Table 7, show that several models – including GBoost and LightGBM – still outperform the regulator's score. As Fig. 3 illustrates, the performance of models using the smaller feature set is comparable to that of models using the full set. This finding suggests diminishing returns from including additional questionnaire items beyond the most informative ones and indicates that a more concise risk assessment may be feasible without a substantial loss of predictive power.

3.5. Interpretation of model performance and practical significance

The R^2s for all models (0.13–0.17) should be interpreted carefully. Although these values might appear low, they are consistent with those in the broader literature on predicting risk preferences. For example, Dohmen et al. (2011) and Beauchamp et al. (2017) find comparably modest explanatory power in their analyses of risk preference elicitation methods.

These consistent findings suggest that predicting financial risk-taking is an inherently challenging task. However, the improvement

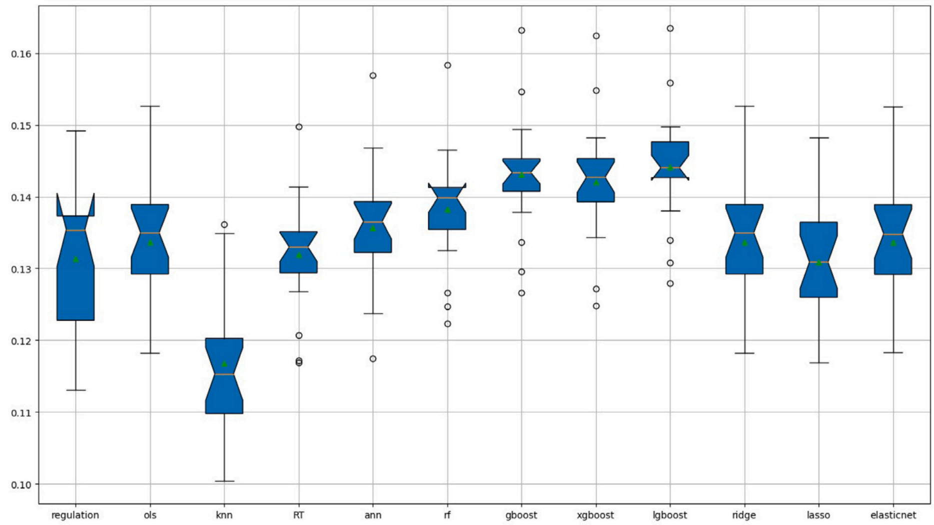

Fig. 2. Model comparison with cross-validation: All features.
Notes: This figure compares the distribution of R^2 across 20 cross-validation folds for every model using the full feature set.

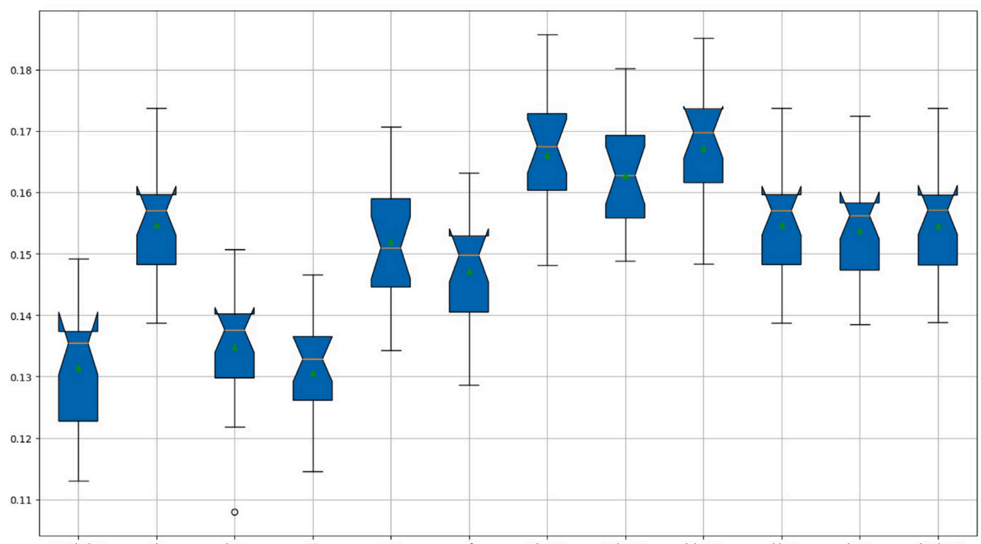

Fig. 3. Model comparison: Selected features.
Notes: This figure compares the distribution of R^2 across 20 cross-validation folds for each model using only the four features selected.

from 0.131 (regulatory benchmark) to 0.167 (LightGBM) represents a 27 percent relative increase in explained variance. In practical terms, this enhancement could translate into more accurate risk categorization for thousands of pension participants, potentially reducing mismatches between recommended and appropriate investment strategies. To contextualize this: correct reclassification by an enhanced model of just 5 percent of participants into more appropriate risk categories could prevent significant portfolio misallocation for approximately 4000 individuals in our sample alone. Extrapolated to Türkiye's entire private pension system, such improvements could benefit a large number of participants.

4. Discussion and conclusion

This study evaluates the effectiveness of machine-learning methods in assessing the risk preferences of 81,563 participants in a pension fund from a major Turkish company (2018–2022). Our analysis contributes to the literature on risk profiling by comparing the predictive performance of various ML models against the regulatory benchmark currently used in the pension industry.

4.1. Discussion of findings

The modest R^2s across all models (0.13–0.17) warrant careful interpretation. Although these values might initially appear low, they are consistent with those in the broader literature on predicting risk preferences. Dohmen et al. (2011) report similar R^2s of 0.10–0.15 in predicting actual investment behavior based on risk preference measures, whereas Beauchamp et al. (2017) find comparably modest explanatory power in their comprehensive analysis of risk preference elicitation methods. These consistent findings across studies suggest that predicting financial risk-taking behavior is an inherently challenging task, influenced by numerous factors beyond those captured in standard questionnaires. Nonetheless, the improvement from 0.131 (regulatory benchmark) to 0.167 (LightGBM) represents a 27 percent relative increase in explanatory power. In practical terms, this enhancement could translate to more accurate risk categorization for thousands of pension participants. Even modest improvements in risk profiling accuracy can have great cumulative effects if the enhanced model correctly reclassifies just 5 percent of participants into more appropriate risk categories. This could prevent significant portfolio misallocation for approximately 4000 individuals in our sample alone.

Our findings both complement and extend recent work in this stream of literature. The analysis by Adekunle et al. (2023) similarly identifies demographic variables, in particular, age, as crucial predictors. However, although their analysis focuses on predicting survey-based risk measures, our approach directly predicts actual portfolio risk, providing a more direct validation of the questionnaire's practical utility. A key factor in this improved performance appears to be the ML models' ability to capture the nonlinear age relationship that we identify (Fig. 1), which adds nuance to their findings and suggests that simple linear models might overlook important life-cycle patterns in risk-taking behavior. Our feature importance results also provide empirical support for the factor structure identified by Kuzubaş and Saltoğlu (2024). Their factor analysis revealed two latent dimensions – risk attitude and financial condition/literacy – and the risk attitude has greater predictive power. Our ML analysis independently confirms this hierarchy, in which Question 3 (self-reported risk attitude) consistently emerges as the most important single predictor. Whereas Kuzubaş and Saltoğlu (2024) report that their two-factor model explains approximately 14 percent of portfolio risk variance, our ML models achieve slightly higher explanatory power (as much as 16.7%), which suggests that the nonlinear relationships and interactions captured by boosting methods offer meaningful, albeit modest, enhancements over linear factor models.

Although boosting methods have the highest predictive accuracy, their relative opacity can be problematic with regard to regulatory acceptance and advisor-client communication, a concern highlighted in the explainable AI (XAI) literature. Our feature selection analysis (Table 5) offers a potential path forward by identifying a stable and interpretable core of four variables that drive most of the predictive power: Question 3 (self-reported risk), Question 4 (lottery choice), age, and age-squared. This finding is particularly relevant, given the operational challenges that pension funds face with lengthy surveys, especially in high-friction channels such as enrollment by phone. A more concise, four-item model, though representing a modest trade-off in predictive power (a 14% reduction over that of the full LightGBM model), is considerably easier to implement at scale. Notably, this simpler model *still* outperforms the original full regulatory benchmark (Tables 6 and 7), which suggests use of a practical balance among accuracy, transparency, and operational feasibility.

4.2. Implications, limitations, and concluding remarks

Our findings lead to several realistic policy and implementation pathways. For instance, a two-level assessment framework could be considered. In such a system, all participants could complete the four-item Core Risk Module (Q3, Q4, age, age-squared) for a rapid and robust initial classification, whereas those who have borderline scores or are close to retirement could be routed to the full assessment or a human advisor. This approach could efficiently allocate resources while addressing implementation challenges. Furthermore, our analysis suggests a re-evaluation of the questionnaire's function. Items with weak predictive power for portfolio risk, such as those related to the investment horizon (Q1) or financial literacy (Q2), could be repurposed as part of a separate advisory and educational module, rather than for risk classification. Finally, our results strongly indicate that any approved risk-profiling model should account for the nonlinear, inverted-U-shaped relationship between age and risk-taking, as simple linear glide paths appear suboptimal.

From an implementation perspective, pension fund companies considering ML approaches should weigh several factors. The initial investment in technical infrastructure and staff training must be balanced against the potential benefits, including better risk categorization and enhanced participant experience through streamlined assessments. Our results show that gradient boosting methods, in particular, LightGBM, offer the most promising performance improvement. However, these gains are modest, and institutions should have realistic expectations about the extent to which ML can improve on current practices. Regular model retraining and validation are essential for maintaining accuracy as market conditions and participant demographics evolve. Furthermore, ethical considerations about the use of demographic variables, especially age, require careful attention to ensure that predictive models inform, rather than determine, investment recommendations.

Several limitations of this study should be acknowledged. First, although our data sample is large and representative of the Turkish pension market, the findings might not be fully generalizable to other regulatory environments or cultural contexts. Second, our analysis focuses on predicting observed portfolio risk, which itself may be influenced by factors other than individual preferences, such as advisor recommendations and default investment options. Third, the overall explanatory power of all the models tested is modest, which suggests that significant determinants of portfolio risk are not captured by standard questionnaires or demographics. The cross-sectional nature of our analysis is another limitation; incorporating a time-series dimension might help identify additional determinants of risk-taking behavior and enable an examination of evolution in risk preferences in response to market conditions.

In conclusion, this study provides empirical evidence on the application of ML techniques to risk profiling in the pension fund industry. Although the modest improvements in predictive accuracy reflect the inherent difficulty of predicting risk-taking behavior, they represent meaningful progress in a critical financial domain that affects millions of people saving for retirement. Our findings suggest that ML methods can enhance risk assessment through better feature selection and nonlinear modeling, but they are not a panacea for addressing the fundamental challenges of preference elicitation. The identification of a small set of highly predictive, interpretable variables offers a potential path toward more efficient risk assessment, balancing the competing demands of accuracy, user experience, and regulatory compliance. Future developments in risk profiling might benefit from hybrid approaches that combine the strengths of traditional questionnaires with ML insights, maintaining transparency while capturing the benefits of advanced analytics. These approaches merit further investigation across diverse cultural and regulatory environments to better understand their potential and limitations.

CRediT authorship contribution statement

Ahmet Göncü: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, and Writing – review & editing. **Tolga U. Kuzubaş:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, and Writing – review & editing. **Burak Saltoğlu:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, and Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.bir.2026.100800.

Data availability

The dataset analyzed in this study is not publicly available due to confidentiality restrictions and proprietary information.

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